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RESEARCH ARTICLE

Effective Underground Facilities Management in Virtual Reality by Applying a Cybersickness Reduction Technique Based on a Four-Degree-of-Freedom Motion Simulator

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ABSTRACT Managing underground utility tunnels (UUT) poses challenges such as safety risks, limited accessibility, and high training costs. To address these, we developed an effective VR-based UUT management training simulator, integrating digital twins with a multi-degree-of-freedom (DoF) motion platform to provide an immersive, cybersickness-reducing experience. Advances in actuator and computer graphics technology have increased the research and development of virtual reality (VR)-based training simulators that use multi-DoF motion platforms. In this study, the proposed system provides precisely synchronized visual feedback through a head-mounted display and kinematic feedback via four-DoF motor-based actuators, realistically simulating UUT conditions, including momentum shifts such as falls and micro-swaying. To mitigate cybersickness, visual and kinematic acceleration control techniques are employed, ensuring users can comfortably and safely experience the UUT without stability concerns. We used a LiDAR sensor to model the actual UUT for realistic visualization in the VR environment and created various scenario-based content, such as facility installation, inspection, and device repair, based on the physical characteristics inside the UUT. This integrated approach enables users to effectively manage and operate UUT facilities in a safe, controlled virtual setting, bridging the gap between traditional training and real-world operations.

INDEX TERMS Motion haptic feedback, motion perception, motion simulator, underground utility tunnel, virtual reality.

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I. INTRODUCTION

With cities growing in size and complexity, considerable infrastructure has been developed within underground utility

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tunnels (UUTs) beneath urban streets. UUTs are used to collectively manage multiple types of urban utilities, such as electricity, telecommunications, water, sewer, and gas, in a single underground corridor. They can increase installation and maintenance efficiency, improve urban aesthetics, and reduce ground settlement and traffic congestion by sharing a common infrastructure path. Therefore, UUTs are essential for sustainable urban development in highly urbanized cities or areas with large-scale development. Depending on the size of the country, UUTs range in length from tens of kilometers to tens of thousands of kilometers. Highly developed cities have UUTs of 2 km per thousand inhabitants [1]. In South Korea, urbanization and industrialization in the 1970s spurred the rapid growth of UUTs, totaling about 500 km as of 2024, starting with Yeouido in Seoul in 1977, to house critical infrastructure such as communication networks and power cables.

As these UUTs age, inadequate ventilation and humidity control contribute to problems like condensation and corrosion. Such conditions can trigger fires, network failures, and power outages, with the potential to paralyze cities and cause substantial economic losses. Consequently, numerous studies have been conducted to ensure the sustainable operation and maintenance of UUTs, with most countries implementing design standards and regular maintenance guidelines [2], [3].

Past incidents associated with condensation and corrosion have underscored the need for effective prevention systems, focusing on improved fire prediction, detection, and response, as well as better environmental management inside the UUTs to reduce equipment malfunctions and downtime [4], [5], [6], [7], [8]. To mitigate these risks, workers must conduct frequent inspections and repairs under challenging conditions, including narrow passages and high electromagnetic fields from high-voltage power cables. Prolonged exposure to such environments may lead to physical and psychological distress, and even severe health issues over time [9], [10], [11]. Therefore, addressing the human factor—ensuring worker safety, training, and preparedness—is just as critical as improving infrastructure itself. However, existing approaches focus on improving disaster response and predictive systems but do not address worker safety and training through immersive solutions.

In this study, we introduce a novel virtual training method for UUTs by integrating digital twins with a four-degree of freedom (4-DoF) motion platform, thereby enhancing immersion and enabling more realistic emergency response and maintenance training. Unlike existing VR training systems, our proposed simulator couples a motion platform, motion recognizer, and high-resolution content while incorporating cybersickness reduction techniques to ensure prolonged user comfort. We leverage threshold-based control—derived from vestibular–visual differences and actual treadmill velocities—to further mitigate cybersickness. This approach is complemented by tracking natural walking patterns via motion capture sensors to estimate and adjust treadmill velocity, ultimately providing a stable and engaging VR experience. We adopted methods from previous studies that identified minimized visual–vestibular gaps as critical for reducing cybersickness and conducted a usability evaluation to comprehensively verify the system's effectiveness. As a result, this solution not only provides a safer and more controlled training environment but also has the potential to advance VR training technology and safety practices in UUT maintenance and operation, ultimately contributing to the broader field of sustainable urban infrastructure management.

This study is organized as follows: Section II reviews previous studies on motion platforms and cybersickness reduction methods. Section III describes the implementation of the proposed motion platform device, motion recognition, and cybersickness reduction techniques. Section IV presents the experimental results, and Section V presents the conclusion and describes the limitations of the proposed device.

II. RELATED WORKS

The proposed VR training systems face two critical challenges: providing naturalistic, immersive motion experiences and mitigating the resulting cybersickness that can arise when visual and vestibular cues are not perfectly aligned. Addressing these issues requires examining both the evolution of motion platforms designed to enhance realism and the methods developed to reduce cybersickness in VR environments.

A. MOTION PLATFORMS

Motion simulation performance is a key factor in the development of realistic VR training simulators. To simulate multiple DoF motion, most motion platforms adopt motion-cueing algorithms (MCAs) based on a hexapod motion system [12], [13], [14], [15].

Because of their geometric configuration and independent control of each actuator, hexapods can precisely control 6-DoF motion, rendering them versatile for applications that require high accuracy, such as flight and vehicle simulations. Furthermore, modern commercial hexapod systems are equipped with advanced control algorithms and user-friendly interfaces that render integration into various simulation settings facile [16], [17]. However, these systems are driven only within a limited translational range in the space of the structure. They cannot simulate walking, running, or other forms of movement, which considerably reduces immersion and realism when used in VR applications such as physical rehabilitation and sports training.

Treadmill simulators are used to simulate walking and running movements in physical rehabilitation, sports training, and military training. Although treadmills that enable users to enjoy physically active VR adventures have been developed [18], [19], [20], they require users to slide across the floor, wearing specially designed shoes with their bodies strapped to the device, resulting in an unnatural walking experience. A high-speed two-dimensional (2D) treadmill with a power transmission mechanism called ''Geared Omni-pulley'' has been devised [21]; however, the system is

bulky and heavy and the expansion of rotational freedom is challenging. Disney developed a treadmill using holotile technology [22]. A holotile floor is composed of numerous disks set at specific angles that enable objects or people on the surface to move and rotate, creating a seamless, interactive experience. However, each disk consists of a complex assembly of gears and motors, rendering the system uneconomical and difficult for commercial use by small companies.

B. CYBERSICKNESS REDUCTION METHODS

Vection is a sensation of illusory self-motion without actual movement and has been studied comprehensively. Visually induced motion sickness (VIMS) resembles conventional motion sickness but occurs without physical movement [23]. It is categorized into subcategories, namely cybersickness in virtual environments and simulator sickness in driving or flight simulators. When individuals are exposed to virtual environments on large or curved screens, they experience cybersickness. This condition is characterized by nausea, oculomotor disturbances, and disorientation caused by sensory discrepancies between visual, somatosensory, and vestibular inputs. These symptoms occur rapidly, affecting 20%–50% of users [24], [25], [26], [27]. The condition generally occurs when visual cues in a virtual environment do not align with the body's somatosensory and vestibular senses. Therefore, posture control based on individual sensory priorities is crucial during VR experience [28]. Recent research has demonstrated that VR-based language education systems can enhance immersion and engagement, thereby supporting the effectiveness of VR in educational training. However, the study did not address specific strategies for mitigating cybersickness [29]. Additionally, Anwar et al. introduced a comprehensive framework for evaluating Quality of Experience (QoE) in immersive media consumption within autonomous vehicles, termed the ''Moving Metaverse.'' The study emphasizes key QoE factors such as presence, immersion, and cybersickness, and discusses the challenges of motion-induced sickness in VR environments. To address these issues, the authors advocate for adaptive content design, network optimization, and the integration of standardized assessment methodologies to minimize motion sickness and enhance user immersion [30].

Sensory conflicts in VR environments are a critical topic of research. Finding effective and intuitive locomotion techniques is a challenge for minimizing fatigue and cybersickness. Moreover, maintaining immersion remains challenging. Studies have analyzed scenarios in which users physically move in one direction in a real environment. However, the study reported that motion sickness still occurs when moving in a different direction because of the sensory mismatch between visual and vestibular stimuli.

Studies on cue conflict theory have investigated the individual factors that induce cybersickness to diagnose and alleviate symptoms in VR environments. A study found a potential method for assessing visual cue sensitivity to variability [31]. The results demonstrated that a higher sensitivity to motion parallax in VR was a significant predictor of motion sickness severity. Studies have investigated object motion in a VR headset to induce moderate levels of motion sickness. After completing the motion task, the participants viewed stereoscopic videos inducing discomfort in the VR headset. The variables of sensitivity to motion sickness induced by VR varied among individuals. Furthermore, observers tended to self-regulate motion sickness through head movements. The results also revealed that sensitivity considerably affected rotational head movement. Although the suppression of head movements was observed during translational motion, the effect was significant during rotational motion. This phenomenon suggests that rotational motion is a major factor that induces motion sickness in VR.

Another study investigated the potential of synchronizing physical movements with visual stimuli to alleviate VR-related discomfort. For instance, a system that uses a motion platform to simulate various levels of visual–vestibular synchronization conditions can reduce cybersickness [32]. The results indicated that users experienced lower levels of cybersickness when their overall comfort improved. This improvement occurred under the synchronized visual–vestibular condition in multiple trials. The provided conditions simulated visual–vestibular synchronization between visual rotation and synchronized visual–vestibular motion to generate basic movements. Furthermore, the study included vestibular motion in a fixed self-referenced visual environment. Minimal sensory adjustment was required when visual and vestibular sensations were aligned with one another. Thus, based on the sensory conflict theory, the severity of cybersickness is likely to be regulated according to individual sensitivity characteristics. The techniques used to synchronize motion platforms could be used to design interactive VR systems. However, developing strategies to minimize cybersickness is crucial in motion platforms driven by voluntary movements.

In early studies, cybersickness was considered less sensitive to the effect of translational motion than to that of rotational motion. However, limited studies have been conducted on generalized reference values for the measurement of cybersickness in VR. A consensus is yet to be reached regarding the specific optical flow and intensity that cause cybersickness. This can be attributed to various reasons; however, limited robust empirical evidence regarding the recommended strategies for VR developers to minimize cybersickness could be a major factor. For this reason, one study generalized the measurement of cybersickness among VR users by distinguishing between their sensitivities to translational and rotational motion [33]. Participants were exposed to various optic flow types in VR to detect rotational and translational velocities and accelerations. The findings indicated that the velocity and acceleration thresholds for rotational optic flow were notably lower than those for translational optic flow. This study demonstrated

the perception of translational velocities on a low-pass filter with different time constants. The field of view (FOV) was assumed to be a cone emitted by an observer. Thus, this study expressed the mean of the transitional velocity ($\bar{v}_{VR}$) of the optic flow in the VR as follows:

$$\bar{v}_{VR} = 2\pi\rho {r_\theta}^2 sv \int_0^h \int_0^{z\tan(\theta)} \frac{r}{\sqrt{\left(r^2+z^2\right)^3}} dr\, dz \quad (1)$$

where θ denote the half angle of the FOV. Furthermore, z denotes the height of the triangle and refers to the distance from the observer's point of view to the location of the virtual object on the z axis. r, r_θ, ρ, h, and s denote the radial distance from the observer, the number of pixels per a half of angle of FOV, the number of particles per physical size (density), the length of the particle field, and the visually perceived size of all particles, respectively. Moreover, based on the actual forward motion velocity ($V_a = 1.2$ m/s), the mean threshold of the perceived transitional velocity was measured to be 1.183, and its standard deviation was 0.836. Therefore, the perceived threshold velocity (V_{VR}) with a 95% confidence interval can be estimated as follows:

$$V_{VR} = 1.183 \pm 0.414, (if\ V_a = 1.2\ \text{m/s}) \quad (2)$$

These thresholds tended to decrease as the particle density in the scene decreased. The average threshold values identified in this study can be useful benchmarks for developing VR applications. To the best of our knowledge, limited experimental studies have focused on strategies for minimizing cybersickness in accordance with the recommended guidelines. Conventional approaches to reducing cybersickness have focused on custom motion settings for individual responses. Methods should reflect the individual characteristics of participants; hence, implementing VR to reduce cybersickness can be complex. In particular, achieving an effective mitigation of cybersickness using conventional strategies when numerous participants are involved is difficult. Studies are required to examine the threshold speed of VR to reduce cybersickness. This could involve applying a generalized speed to mitigate cybersickness.

Studies have revealed that cybersickness can be reduced through behavioral strategies related to an individual's posture depending on the choice of equipment. For example, a study [34] revealed the effects of various motion stimuli combinations and the user's range of neck motion through two experiments. Several navigation tasks were conducted to measure the simulator sickness using four types of VR locomotive controllers. A strategy to reduce cybersickness can be adopted depending on the equipment selection. This proves that spontaneous posture considerably affects cybersickness reduction. In some cases, the predictive abilities of conventional and recurrence quantification analysis-based measures of postural activity were compared by investigating spontaneous fluctuations in foot pressure in a stationary position [35]. When exposed to an optical flow, participants who displayed lower foot pressure were more likely to report stronger vection and VIMS. These results indicate that individual differences in spontaneous postural stability enable the prediction of the likelihood of experiencing VIMS in the future. This study demonstrated that individuals with greater postural instability while standing still in stationary visual environments are more likely to experience cybersickness. The study reported that this effect occurred after exposure to visual motion stimulation. However, addressing cybersickness when considering subtle individual differences and their impact could be less feasible. Establishing a validation method that represents reliable cybersickness reduction is necessary.

Various studies have utilized external peripheral devices to address this issue by devising different methods to reduce sensory conflict and enhance the user experience [36], [37]. These methods involve tracking the user's physical movements and translating them into virtual locomotion. Exploring newer approaches that incorporate external devices or simulators capable of reflecting a user's physical movements offers a promising solution for mitigating cybersickness [36]. Moreover, effectively synchronizing an individual's active movements in strategies designed to minimize the induction of cybersickness by incorporating natural walking characteristics is essential [37]. Studies have focused on exploring and designing optimized methods for natural walking velocity adjustment. A controlled treadmill enables users to adjust their walking velocity; however, this experience differs from the experience of walking on actual ground. Overcoming the sensory mismatch between visual and vestibular stimuli is crucial for reducing disparities between treadmill and real-world walking environments and is essential for ensuring an accurate simulation of real-world walking conditions during treadmill exercise. Studies on walking in VR synchronized with a treadmill have continuously investigated the design and analysis of systems that update the walking velocity. A study considered various aspects of walking motion characteristics to create a natural walking environment [38]. In the study, two locomotion techniques, an omnidirectional treadmill and arm swinging, were compared in the context of VR environments. According to this study, although the treadmill was designed to simulate walking, users did not feel as if they were walking, particularly during directional changes. Several problems regarding methods that consider walking motion characteristics remain unresolved.

The accurate estimation of treadmill control speed remains a significant challenge in treadmill research. Studies have established that this estimation is vital for achieving seamless integration of VR experiences with natural walking, considerably improving the participants' sense of immersion. Consequently, many studies have focused on minimizing the discrepancies between natural walking and visually induced VR movements to reduce the risk of cybersickness [39], [40], [41], [42], [43], [44]. Studies used a treadmill velocity update system based on the swing phase, reflecting walking velocity, have been conducted. For example, [45] proposed a

feedforward control method that estimates walking velocity based on the maximum swing foot velocity in the swing phase. In experiments involving patients with Parkinson's and freezing of gait, a system was used to provide a safe and realistic walking environment. According to the study, the walking velocity was obtained from the maximum velocity of the swing phase (V_{max}). The velocity of the swing foot is (V_{foot}):

$$V_{foot} = V_{max} \sin\left(\frac{\pi}{T}t\right) \tag{3}$$

where T denotes the swing time of one gait cycle in the domain of time t. Foot speed was not uniform during walking. According to (3), the swing foot travels through the first half of the sine function period in the swing phase (from toe-off to heel-strike). The study generalized the velocity of the swing foot based on kinematic features; hence, the mean velocity of the pelvis could be estimated. To estimate the walking speed, in this study, the distance traveled by the feet and pelvis was measured. The swing foot distance along the walking direction was estimated to be two times the distance traveled by the pelvis. The mean walking velocity with respect to the pelvic position ($\overline{V}_p$) is represented as follows:

$$\overline{V}_p = \frac{1}{2T}\int_0^T V_{foot}(t)dt \tag{4}$$

The treadmill generated a belt velocity facing the direction of walking equivalent to the mean velocity of the pelvis obtained in (4). Consequently, the treadmill belt velocity was updated using the estimated mean velocity of the pelvis in the middle of each swing phase. This approach achieved faster response times than that of other gait parameters. It enables users to control their walking velocity on a treadmill, providing an environment similar to that of actual walking. The proposed treadmill control system also ensured stability for walking to keep users close to the center of the treadmill track. By contrast, based on other gait analysis studies, the terminal stance (heel-off) phase has been widely studied as a critical state that generates propulsion, considerably affecting the walking velocity [46]. The study revealed that the swing phase constituted approximately 40% of the gait cycle, with the least muscle activation. For optimal treadmill velocity adjustments, gait measurements should focus on the final stance phase rather than on the swing phase. Given these points, the timing of speed adjustments remains uncertain, indicating that using foot velocity alone is challenging. To overcome the limitations of swing foot tracking, exploring simple and precise methods is essential. Using the average pelvic velocity to control the treadmill speed could be beneficial for stable walking, regardless of specific timing updates.

Another study used an omnidirectional treadmill that enabled natural walking to alleviate cybersickness. According to [47], an omnidirectional treadmill replicates a user's physical movements in a virtual environment, enabling unaffected natural locomotion. In terms of user experience,

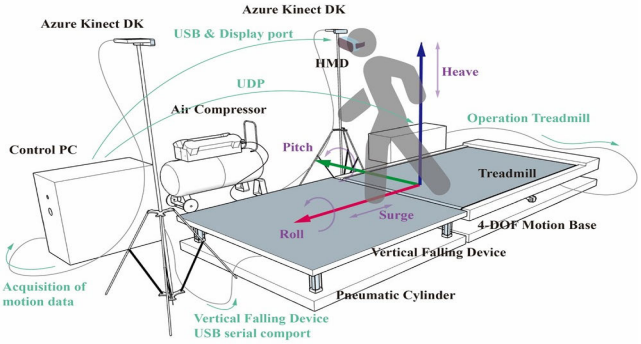


FIGURE 1. Structure of the proposed underground utility tunnel (UUT) trainingsimulator.

the treadmill outperforms conventional game controllers. Despite enabling a natural gait on a treadmill, the technology could increase user fatigue and induce cybersickness. Further studies are necessary to enhance user experience and immersion. This result supports those of prior studies, which indicate that individuals tend to experience less cybersickness during linear motion.

Anwar et al. investigated the impact of motion-related factors, such as camera movements (fixed, horizontal, vertical) and the number of moving targets, on cybersickness and QoE in 360-degree VR videos. Their findings highlight that vertical camera motion and multiple moving targets significantly increase cybersickness, assessed using the Simulator Sickness Questionnaire (SSQ) [48]. However, the study focused solely on visual motion and lacked a physical motion platform, limiting its ability to address visual-vestibular conflicts effectively.

Thus, previous studies have confirmed that cybersickness induced by rotational motion is more sensitively affected by VR than that by translational motion. Further studies are necessary to investigate the effects of a feasible VR controller with simulated perceived transitional movements.

The majority of existing approaches emphasize certain factors—such as rotational motion-induced discomfort or constrained locomotive capabilities—without providing comprehensive, real-time adjustments to accommodate individual differences or maintain continuous sensory alignment. In contrast, the approach described in this paper integrates a 4-DoF motion platform, real-time walking velocity estimation, threshold-based cybersickness reduction strategies, and vestibular–visual synchronization techniques. By doing so, our method directly addresses the collective shortcomings highlighted in the literature, offering a more natural, responsive, and comfortable VR experience for diverse users. In this study, we hypothesized significant main effects of Age group and Condition on the multivariate means of Eye Strain, Headaches, and Dizziness, and further posited an interaction between these variables. Specifically, we proposed that the effect of one independent variable would vary depending on the level of the other. We tested these hypotheses via

MANOVA, and the findings supported the hypothesized reduction in cybersickness across the respective groups.

III. DESIGN AND IMPLEMENTATIONS

A. UUT TRAINING SIMULATOR

The proposed UUT training simulator was composed of a control PC, a 4-DOF motion platform, a vertical fall device, a motion recognizer, and an HMD (Fig. 1). The virtual reality content of the UUT for experiential training is modeled using high-precision LiDAR [49], which uses Simultaneous Localization and Mapping (SLAM) technology to automatically perform registration on the move, and is designed to be similar to the actual tunnel so that users can experience various facilities in the UUT, such as communication lines, power lines, and water supply (general water and industrial water), to satisfy the requirements of experiential training. When the system initiates, the virtual reality content installed on the PC is visualized on the HMD, and the motion recognizer recognizes the user's three-dimensional posture and position in real time.

The motion recognizer used in the system incorporated an Azure Kinect DK [50] and a Time of Flight (TOF)-based motion recognition sensor to acquire the joint position and motion data of the human participants. Using its built-in body tracking SDK, the Azure Kinect DK estimates a 3D skeletal model comprising 25 joints mapped to the human body structure in real time, based on depth data and machine learning algorithms.

The motion during walking, centered on the base of the spine of the tracked joint skeleton, was estimated in real time and applied to the virtual training content of the UUT. The forward direction and speed of the participant relative to the base of the spine determined the surge direction and speed of the motion base. It constrained the user to stay centered and not fall off the motion platform.

The motion base and PC were wired through UDP and USB communication with the vertical falling device. The motion base was precisely synchronized and simulated according to the terrain in the virtual tunnel (inclination angle on the x- and y-axes, fall, etc.) so that the user can experience realistic UUT training within a limited space.

B. MOTION SIMULATOR

The proposed motion simulator consisted of a motion platform and a vertical falling device for controlling a 4-DOF (roll, pitch, surge, and heave) motion. The nature and purpose of the motion feedback of these two devices in the simulator differ considerably; therefore, they are operated independently using separate controllers for ease of implementation.

The motion platform has 4-DoF and is categorized into upper and lower driving parts. The bottom driving part is called the motion base and is powered by three motorized units (A, B, and C), as depicted in Fig. 2, providing 3-DoF motion (roll, pitch, and heave).

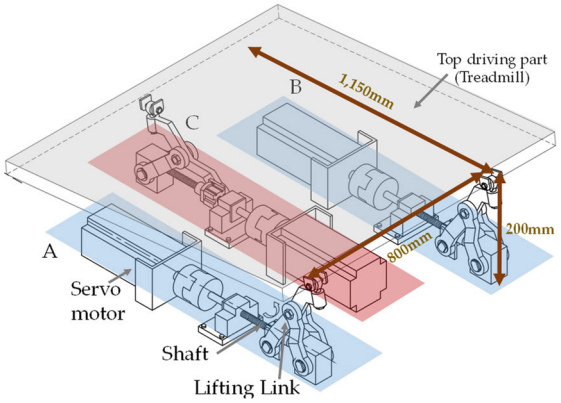


FIGURE 2. Proposed 3-DoF motion base.

Each driving unit consists of a servomotor, a shaft, a lifting link, and devices for fixing the parts. The links installed in driving units A and B were in the same direction. The link installed in driving unit C was in the opposite direction, and these three links were connected to the top driving unit. In this case, the axial coupling direction between the lifting link of driving unit C and the top driving unit was perpendicular to the axial coupling direction between the lifting link of driving units A and B and the top driving unit. That is, the axis coupled between the lifting link of driving unit C and the top driving unit is parallel to the shaft. With this configuration of driving units, the bottom drive unit performed pitch, roll, and heave. The top driving part, also known as the treadmill simulator, consisted of a servomotor, rubber belts, shafts, and a walking pad to perform the surge of the motion base (Fig. 3). These structures created a smooth motion, providing users with a realistic and smooth walking experience in the VR space.

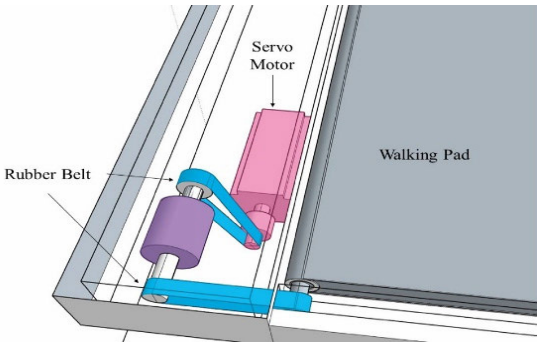


FIGURE 3. Proposed top driving unit (treadmill).

The vertical falling device uses four pneumatic cylinders to make the motion platform perform other heave motions independent of the motion base, as depicted in Fig. 4, to provide the user with a realistic falling experience. Each cylinder was connected to an individual 5/2 solenoid valve, which received an electrical signal from the controller and air from the compressor to control the flow direction and speed of the air, causing the plate to rise or fall. The air cylinder

used in the implementation had a stroke length of 30 cm. The maximum operating speed of the cylinder was approximately 2 m/s. Therefore, the piston rod required 0.15 s to move, which is faster than the speed at which a stationary person falls in free fall (0.25 s).

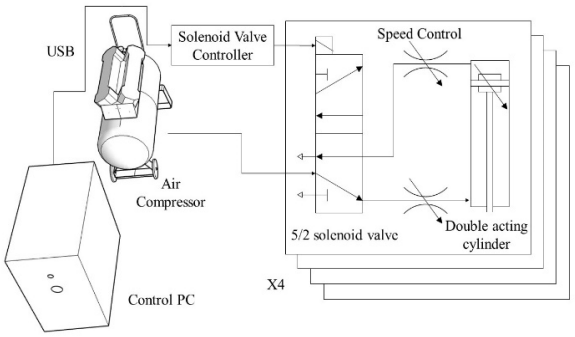


FIGURE 4. Vertical falling device.

C. POSE ESTIMATION AND CYBERSICKNESS REDUCTION

The proposed system incorporated an Azure Kinect sensor to capture the walking motion on a treadmill. The sensor measured the reference position, which was set at the center of the treadmill (COT). Similarly, the sensor acquired the pelvic joint and spine base from skeletal data in the forward walking direction at the COT. When tracking the pelvis and coordinates using Kinect, the control PC profiles each velocity by coordinating to generate an adaptive velocity, which determines the control speed of the treadmill. A given set of coordinates represents the motion data in real time according to the position and direction of the skeletal joint (Fig. 5).

The system calculates the participants' positions on the 1.5-m-long treadmill. Measuring gait intervals is essential to guarantee that walking does not sacrifice stability. As illustrated in Fig. 6, S_{ref} is the distance from the COT to the Kinect, and S_p is the distance from the subject's pelvis joint to the Kinect obtained in real time. Equation (5) represents d, which is the range of safe distances to travel, as follows:

$$\left|S_p - S_{ref}\right| \leq d \tag{5}$$

When the motion data captured by the sensor were transmitted to the control PC, the input data were used to estimate the speed of the walking user and calculate the treadmill velocity control. The swing phase velocity per half-foot stride period is comparable with the translational velocity of the periodic pelvis on the treadmill. To control the speed in the parallel directions in the VR, we used the speed values in (6). For walking velocity, the translational distance per second (t) and transitional velocity with respect to the pelvis (v_p) can be expressed as follows:

$$v_p = \frac{S_p - S_{ref}}{\left|S_p - S_{ref}\right|} \frac{S_p - S_{ref}}{t} \tag{6}$$

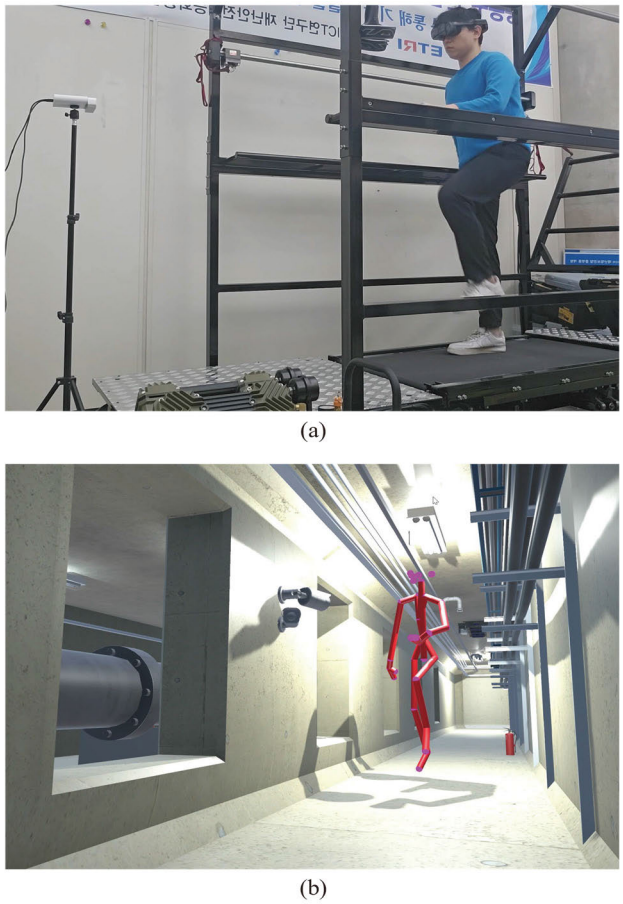


FIGURE 5. Full body pose estimation: (a) hardware setup with a ToF-based motioncapture sensor and 4-DOF motion platform, (b) visualization of posturetracking.

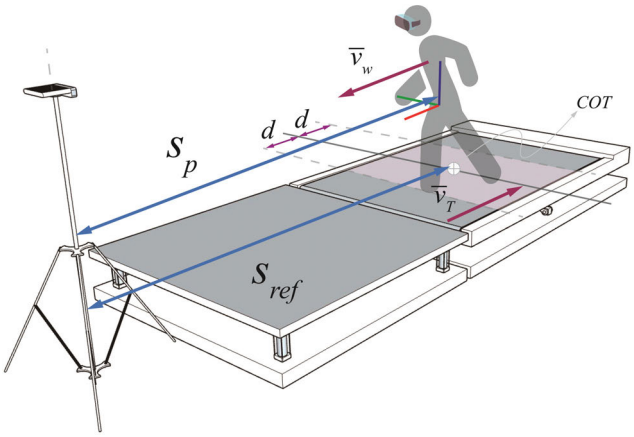


FIGURE 6. Variables for controlling walking (v_W) and treadmill velocity (v_T).

where $\frac{S_p - S_{ref}}{\left|S_p - S_{ref}\right|}$ refers to the variable for the change in the participant's walking direction based on the COT position. Here, $\frac{S_p - S_{ref}}{\left|S_p - S_{ref}\right|} < 0$ indicates the forward walking movement in the sensor direction. The actual walking distance is directly proportional to the movement of the participant's pelvis;

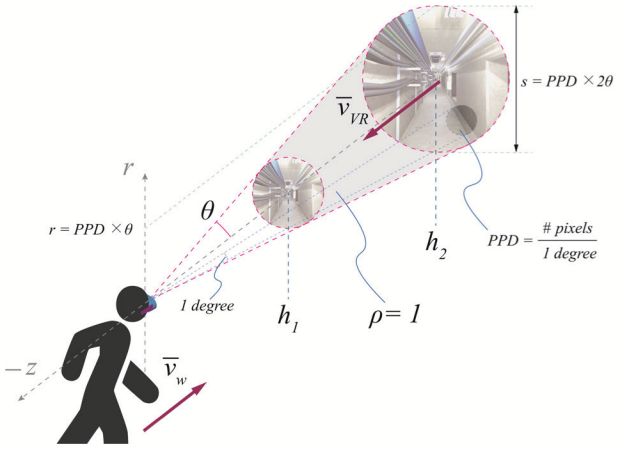


FIGURE 7. Variables for controlling VR transitional velocity (v_{VR}).

hence, the estimated mean pelvic velocity ($\overline{v}_p$) is proportional to the mean actual walking velocity ($\overline{v}_w$).

$$\overline{v}_w = \overline{v}_p \tag{7}$$

where $\overline{v}_w$ is the mean of the actual pelvic travel distance per second. We investigated the actual walking from the measured pelvic velocity to estimate the velocities used in the proposed system, as depicted in (7). According to (4) from previous studies, the pelvic velocity in the walking direction was modeled based on the sinusoidal relationship of the velocity of the swing foot. The pelvic velocity was determined by integrating the swing foot velocity over T s, dividing it by T, and taking half of that value. In simple terms, the pelvic velocity is half the mean swing foot velocity. The treadmill belt should move in the direction opposite to the participant's pelvic movements to ensure a natural walking experience. That is, the mean treadmill belt velocity was controlled to counteract the mean half-foot velocity. By estimating the mean actual walking velocity, the mean treadmill velocity could be adjusted to ensure an unaffected walking experience. Thus, we quantified the treadmill belt velocity and mean foot velocity ($\overline{v}_{foot}$) based on their relationship with the mean of the actual walking velocity as follows:

$$\overline{v}_T = -\frac{1}{2}\overline{v}_{foot} = -\overline{v}_w = -\overline{v}_p \tag{8}$$

$\overline{v}_{foot}$ represents the mean of foot velocity; hence, the proposed motion platform estimates $\overline{v}_p$ at approximately 0.12-s intervals to calculate half of $\overline{v}_{foot}$ according to (8). This value was then used to compute $\overline{v}_T$, which was used to control the treadmill belt.

In the proposed VR training, the participants wore the HTC Vive XR Elite for VR HMD. The resolution of the HMD was 1920 × 1920, which was applied as a parameter to a function that obtained the VR translational motion in the proposed system. According to [33], because it is assumed to be conical in shape, the area that the participants view from the front of the FOV is assumed to be circular. Fig. 7 reveals a view frustum from the observer's perspective and the overall variables of the proposed VR training. Based on (1), the mean of visually perceived translational velocity ($\overline{v}_{VR}$) can be expressed as follows:

$$\overline{v}_{VR} = -2\pi\rho {r_\theta}^2 s \overline{v}_w \int_{h_1}^{h_2}\int_0^{z\tan(\theta)} \frac{r}{\sqrt{\left(r^2+z^2\right)^3}} dr\, dz \tag{9}$$

where z denotes a variable corresponding to the cone height of the view frustum in the direction of the translational motion of the observer. For the volume integral, the z length repressed the cone height from the position of the near-clipping plane h_1 to the position of the far-clipping plane h_2. According to (1), the position of the near-clipping plane of the ideal-view frustum is zero. However, $h_1 = 0$ converges to infinity, resulting in an infinite length of the view frustum. Therefore, from a practical perspective, the position of the near-clipping plane approaches zero rather than precisely substituting for zero. In our VR training, the positions of the near and far clipping planes were calculated as $h_1 = 3$ px and $h_2 = 40,000$ px. By contrast, the integral of the value with respect to a radius of a view frustum on an interval 0 to $z\ tan(\theta)$ represents the size of the visually perceived scene facing the VR observer. The value of θ was calculated to be 55°. According to [33], this value is the angle of half the FOV. The VR FOV of the proposed system was applied at 110°. Considering the radial distance (r), we determined the value within the boundaries of the frustum view. Here, $\overline{v}_{VR}$ can be calculated by simplifying equation (9) and is equal to the constant multiple of $\overline{v}_w$. Furthermore, r_θ denotes the number of pixels per half angle of FOV. Therefore, r_θ can be estimated by multiplying the average pixel density (px/degree), defined as the number of pixels displayed per degree of observer's vision, by half of the rendered FOV angle. With regards to s, the value refers the sum of evenly constant pixels that can be observed. We estimated s by multiplying the average pixel density with the rendered FOV angle. Furthermore, as the virtual training content has a fairly high density of virtual objects, ρ was set to 1 according to methodology implemented in [33]. The maximum values of the z and r were determined. Because the value for the double integral, ρ and s can be computed, we determined a constant value of $\overline{v}_{VR}$. Thus, we can simplify $\overline{v}_{VR}$ as follows:

$$\overline{v}_{VR} = -2\pi\rho {r_\theta}^2 s\,(1-cos\theta)\ \ (ln\,(h_2) - ln\,(h_1))\,\overline{v}_w \tag{10}$$

Visually perceived translation is opposite to the direction of walking; hence, v_{VR} is applied in the negative direction of walking velocity. For unit correction, the values of s h_1, h_2, and r were converted to 0.00026 m per pixel.

To investigate cybersickness, we need to determine the physical size of the gap between the visual and vestibular stimuli that cause vection. As expressed in (2), a previous study investigated the 95% confidence interval of the threshold velocity at which cybersickness was not induced in visually perceived motion in VR. The gap

between the visual and vestibular stimuli is related to the difference between the visually perceived velocity threshold and actual speed ($V_{VR} - V_a$). Similarly, the proposed method obtains the mean walking velocity ($\bar{v}_w$) performed simultaneously by using a motion capture sensor, as presented in (7). Additionally, we estimated the mean visually perceived transitional velocity ($\bar{v}_{VR}$) during UUT VR training, as expressed in (10). Thus, we represented the velocity differences between the visual and vestibular stimuli (v_D). Based on (8), we assumed the difference between the visually perceived and walking velocities in the UUT VR training as follows:

$$v_D = V_{VR} - V_a \simeq \bar{v}_{VR} - \bar{v}_w \simeq \bar{v}_{VR} - \bar{v}_T \tag{11}$$

To prevent cybersickness, the velocity differences between the visual and vestibular stimuli can be calculated using (2) when the actual walking speed is 1.2 m/s. The cybersickness-free velocity range can be expressed as follows:

$$v_D \in R, (R = [-0.431, 0.397]) \tag{12}$$

Thus, we propose a method to determine the minimization of cybersickness in our system by the physical quantity for v_D, which implies velocity differences between visual-vestibular stimuli. This indicates that the closer the value is to zero, the less the induced cybersickness is, according to (11). When the participants walked on a motion platform and performed VR UUT training, the proposed system recorded the translational motion speed of the presented VR every 0.03 s. We measured cybersickness performance from the range of error between treadmill-presented translational velocities using (12).

The proposed system matches the real walking velocity and visually perceived motion speed in VR by applying a motion capture sensor to reduce the effect of cybersickness and improve immersion. Fig. 8 depicts the overall control structure of the proposed system.

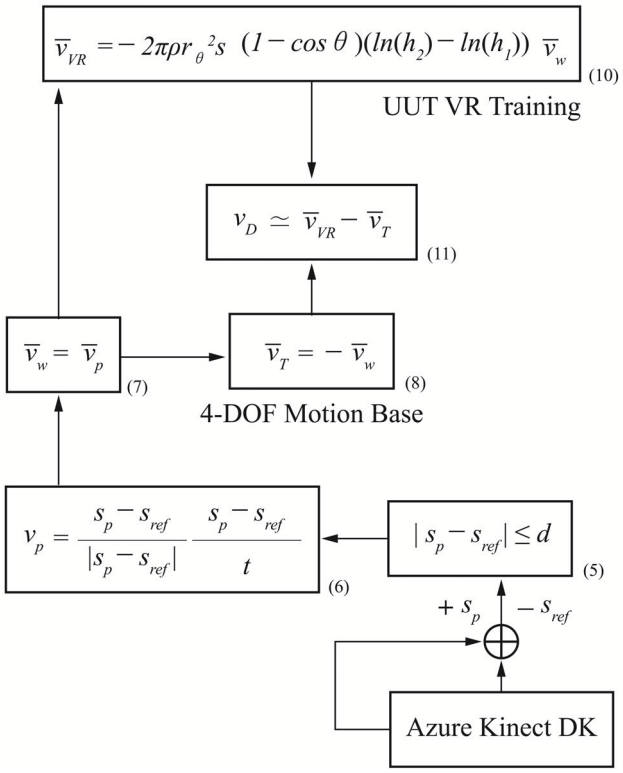


FIGURE 8. Control structure of proposed system.

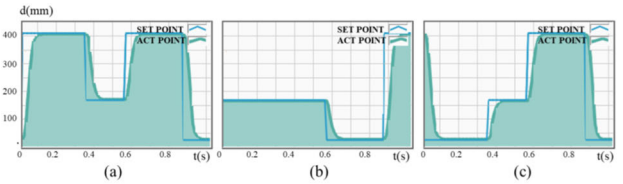


FIGURE 9. Response latency of the 3-DOF motion base: (a, b, c) travel distanceof the shaft for each driving unit in a second.

IV. EXPERIMENTAL RESULTS

The proposed device can best be understood through detailed experimental results and use cases of the UUT virtual training platform with the cybersickness reduction technique as described in the following sub-sections.

A. MOTION PLATFORM

All motors used in the motion base are ECMA-C10807ES from Delta, controlled by a motion control card (PCI-DMC-F02) with a PCI interface. Table 1 presents the specifications of the motion base. The speed of the 4-DoF motion platform was determined by the specifications of the motor and the number of threads grooved into the shaft.

The motor used in the implementation had a rotational speed of 5,000 r/min, and the 400-mm long shaft had 10 threads. Therefore, the time required for the lifting link to actuate is 0.12 s, as depicted in Fig. 9. The graph reveals the actual shaft position movement versus the set shaft position for each driving unit (a, b, and c) of the developed 3-DOF motion base, indicating the driving latency and motion performance of the motion base.

The ideal motion speed of the pitch is ±41.7°/s. However, the speed and acceleration of the motor decreases as the weight of the participant and the top drive (treadmill) increases. Therefore, the final angular velocity was measured to be ±35°/s. The angular velocity was estimated by using the motor speed obtained from the SDK of the motion card. The heave range of the falling device was 300 mm depending on the stroke length of the cylinder, and the heave range of the motion base was 200 mm, depending on the height of the lifting link, as depicted in Fig. 2. Furthermore, the rigid wheel base of lifting links AB and C in Fig. 2 is 1,150 mm, and the distance between lifting links A and B is 800 mm. Therefore, the driving range of roll and pitch were ±14° and ±10°, respectively.

TABLE 1. Detailed specifications of the motion platform.

Operating range	Roll	±14°
	Pitch	±10°
	Heave (Vertical fall device)	300mm
	Heave (Motion base)	200mm
	Surge	1,500mm
Operating speed	Roll	±40°/s
	Pitch	±35°/s
	Heave (Vertical fall device)	2.0m/s
	Heave (Motion base)	1.5m/s
	Surge	10km/h
Payload		400kg

B. PERFORMANCE OF THE CYBERSICKNESS REDUCTION

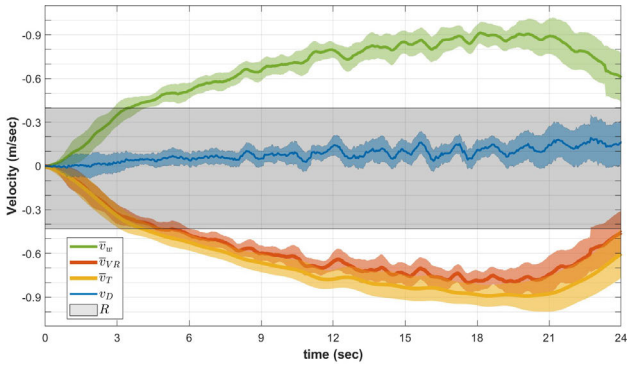


FIGURE 10. Cybersickness results: actual walking ($\bar{v}_w$, green), the average of velocity of the VR ($\bar{v}_{VR}$, red), the average of velocity of the treadmill ($\bar{v}_T$, yellow), difference between the visually perceived and walking velocities (v_D, blue) and cybersickness-free velocity range ($R = [-0.431,\ 0.397]$, grey area).

In this section, we observed real-time values for all velocity and pelvis positions at 0.03 s intervals when the participants walked. After completing VR training when walking, a user test was performed. If a participant complained of cybersickness, all tasks were stopped immediately. As depicted in Fig. 10, we measured the average distribution of the velocities corresponding to the pelvic position on the treadmill. We also measured actual walking movements and visually perceived translational movements in VR. Finally, we calculated the difference between the VR movements and the actual walking movement in m/s. Equation (2), as presented in previous studies, is the confidence interval of the threshold speed for preventing cybersickness. If the velocity was within the confidence interval, the difference between the visual and vestibular stimuli was not notable. Therefore, cybersickness was not induced. This result demonstrated that the cybersickness effects by measuring two physical phenomena in which different inducing visual and vestibular stimuli are synchronized with VR and actual reality ($V_{VR} - V_a$).

We examined whether the proposed VR velocity was within a threshold velocity that did not induce cybersickness, as depicted in (11). As shown, the threshold interval ($v_{VR} - v_w$) was -0.431 to 0.397, representing the error between the actual and virtual velocities. As illustrated in Fig. 10, we observed a threshold interval within the range required to prevent cybersickness, as suggested in studies. The results indicated that the walking velocity within the proposed system remained within the threshold. Thus, we demonstrated that the proposed VR training system facilitated limited difference between the visual and vestibular stimuli, effectively preventing cybersickness. We ensured that the proposed system provided an immersive VR experience by minimizing sensory collisions between physical and virtual sensations. Improving the accuracy and precision of the walking motion can result in efficient and safe movements.

For the experimental tasks in the proposed system, all participants ride on the presented treadmill and subsequently wear an HMD to begin VR UUT training. When the participants entered the VR tunnel stage, they paused for a moment according to the instructions for the training, as depicted in Fig. 11. Furthermore, our system began calibration using a motion capture sensor. If the participant was far from the center of the treadmill, then it was guided to the center and adjusted. The sensor identified the participant as being in the central position and, after a few seconds, entered the VR tunnel stage. When the participant started walking, the treadmill velocity was adjusted according to the walking position and movement direction. As the participant moved forward and backward from the center of the treadmill, the treadmill velocity gradually increased and generated a speed such that the participant was as close to the center of the treadmill as possible.

C. USER TEST

We applied the proposed simulator to a VR training application that exercised the use of an underground facilities management. Fig. 11 depicts the UUT VR training from the participants' perspective and an overview of the external model. In this user test, the user performed facility management and inspection inside an actual underground tunnel for a long time and subsequently compared the training experience by applying various scenarios that could occur inside the UUT, such as walking in the VR tunnel, inspecting communication cables and power cables, and extinguishing fires. During the VR training, various visual and motion feedback was provided depending on the tunnel terrain: flat, uphill, downhill, and falling.

For user testing, 17 public participants (10 men, 7 women, ages 20–50) who had no experience with VR

TABLE 2. Analysis of cybersickness symptoms and intensity by age experienced by participants with and without the cybersickness technique.

Age Group	Number of participants	Without cybersickness reduction technique			With cybersickness reduction technique		
		Eye strain (1-10)	Headaches (1-10)	Dizziness (1-10)	Eye strain (1-10)	Headaches (1-10)	Dizziness (1-10)
20s	4	5.0	3.5	5.0	5.0	3.0	4.5
30s	5	4.8	3.0	4.0	3.8	2.4	4.0
40s	4	5.5	4.0	6.5	5.5	3.5	5.0
50s	4	7.0	5.5	8.0	7.0	4.0	5.5

applications based on motion platforms participated in the training.

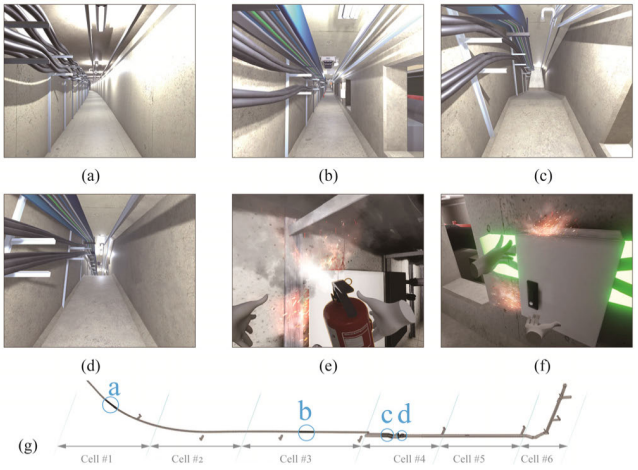


FIGURE 11. VR UUT content model: (a) UUT VR Training of participant's viewpoint of curve passage, (b) straightpassage, (c) downhill passage, (d) uphill passage, (e) and (f) performing UUT management by VR Training, and (g) a model of theentire aisle inside the UUT from the side.

Also, the participants were given detailed explanation of the purpose of the research the experimental procedure, and they provided informed consent. Prior to training, they were trained to walk over 2 km in a real underground tunnel for 1 h, inspect communication and power cables, and then repeat similar training in a virtual environment for 20 min.

Fig. 12 shows the results of a survey evaluating motion feedback, VR immersion, and cybersickness compared to walking in a real underground tunnel, presented on a radar chart. The survey included the Presence Questionnaire (PQ) to assess user experience and immersion, using a five-point Likert scale where higher numbers indicated greater immersion and similarity to the virtual environment. In addition to the high-quality, photorealistic virtual UUT content created in Unreal, the implemented motion platform accurately simulated the slope of the virtual road surface on which the participant was walking, as illustrated in Fig. 11; thus, the participants felt that the visual and motion feedback was realistic.

Table 2 presents the symptoms and intensity of cybersickness experienced by participants, averaged by age. By administering the SSQ before and after VR sessions, we asked participants to rate their levels of eye strain, headache, and motion sickness after experiencing one simulator with and without cybersickness reduction techniques. We employed the SSQ, originally introduced by [51], to collect participants' validated self-reported responses regarding cybersickness. Through the MANOVA analysis, we identified the significant effects of the the proposed cybersickness reduction method and age groups on cybersickness symptoms, including eye strain, headaches, and dizziness. First, the overall model demonstrated a strong fit to the data, with significant explanatory power for the dependent variables. Specifically, Wilks' Lambda was 0.0608, with an associated F-statistic of 138.97 ($p < 0.001$), indicating that the combination of the independent variables (Age Group and Condition) significantly influenced the cybersickness scores. Regarding the effect of Age Group, the analysis showed a statistically significant impact on cybersickness scores, with Wilks' Lambda $= 0.2355$, $F(9, 65.86) = 5.94$, $p < 0.001$. This result suggests that experiences of cybersickness symptoms vary significantly among different age groups. Additional metrics, including Pillai's Trace and Hotelling-Lawley Trace, corroborated this finding, reinforcing the robustness of the age-related differences.

Participants in their 30s experienced the least eye strain, headaches, and motion sickness, whereas those in their 50s experienced the most strain. The participants experienced the same eye strain with and without the cybersickness reduction techniques; however, headaches and motion sickness were less severe with the cybersickness reduction techniques. In particular, participants in their 50s experienced a considerable reduction in the intensity of their headaches and motion sickness. Older adults have a higher acceleration threshold at which they begin to detect actual physical surge movements [52] and are more comfortable walking on a treadmill that accelerates and decelerates than younger participants. Standard deviations ranged up to 1.4, indicating moderate variability across subgroups. The normality tests using Mardia Skewness ($p = 0.6717$) and Mardia Kurtosis ($p = 0.9244$) indicated no significant deviation from a multivariate normal distribution. The condition variable, which compared the presence or absence of the cybersickness reduction technology, also had a significant effect on the dependent variables. Wilks' Lambda for this factor was 0.6780, with $F(3, 27) = 4.27$, $p = 0.0136$.

However, the age group × condition interaction was non-significant (Wilks' $\lambda = 0.75$, $F(9, 58.56) = 0.81$, $p = 0.61$). Thus, in this study, age group and condition independently influenced eye strain, headaches, and dizziness, while their combined effect did not reach statistical significance. These results indicate that the cybersickness reduction technology effectively reduced symptoms of cybersickness, leading to lower scores for eye strain, headaches, and dizziness when compared to the traditional condition without the technology.

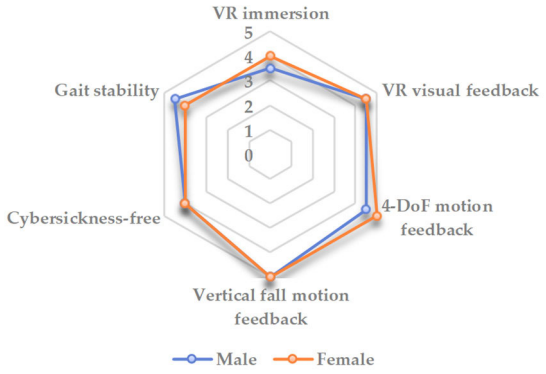


FIGURE 12. Radar chart (Gait stability, cybersickness-free, vertical fall motionfeedback, 4-DoF motion feedback, and VR visual feedback, and VRimmersion) surveyed from user experiences.

V. CONCLUSION

This study provided a realistic UUT training system with minimal motion sickness to participants by precisely controlling visual and physical object movements in a virtual environment using a 4-DoF motion platform. The proposed system provides realistic visual and various kinesthetic motion feedback, such as sudden fall while walking and micro-vibration caused by trucks passing by on the ground, by closely examining the real UUT environment and modeling it using a LiDAR sensor. Furthermore, visual and kinesthetic acceleration control methods were applied to minimize motion sickness in participants experiencing VR on a motion platform to provide a comfortable experience training for participants. To evaluate the proposed simulator, we analyzed the cybersickness induced by experiencing the UUT training program with 17 participants of various ages. The results revealed that compared with existing VR solutions, the training not only reduced dizziness and motion sickness but also demonstrated the effectiveness of facility management training in a UUT.

The proposed motion platform should be improved considerably. Because most of the interior of the UUT is a long straight passageway, the 4-DoF motion platform proposed in this study is sufficient for simulating such a tunnel. However, to satisfy the various types of VR applications, supporting yaw or sway motions is difficult. As the system has 4 DOFs, we are currently working toward adding yaw to create a 5-DOF motion platform to realize an unlimited yaw movement capability of 360° through a rotary drive on top of the motion base [53].

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